

ORIGINAL RESEARCH

Enhancing Biofilm Recovery From Carbon Filters for Molecular Analysis and Flow Cytometry

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Received: 31 March 2025 | **Revised:** 30 June 2025 | **Accepted:** 11 July 2025

Deputy Editor: Lauren Weinrich | **Associate Editor:** Leigh Gilmore Terry

Funding: This work was supported by the National Science Foundation (1832713).

Keywords: biofilm | carbon filters | flow cytometry (FCM) | molecular analysis

ABSTRACT

Effective biofilm recovery is essential when characterizing microbial communities on granular filter media in water treatment applications. This study evaluated (1) DNA extraction kit modifications to maximize DNA recovery and (2) low frequency sonication (40 kHz) durations to maximize recovery of viable cells for flow cytometry (FCM) considering filter media type (anthracite and granular activated carbon [GAC]), media age, and filter runtime. Both DNA and intact cell concentrations were greater for new GAC (< 12 months of full-scale operation) and anthracite compared to old GAC (> 20 years). Biomass yields for old GAC were improved by using a less aggressive bead-beating protocol (for DNA) and reduced sonication duration (for intact cells). Finally, collecting media at the beginning of a filter run (i.e., post-backwash) improved biomass recovery compared to sampling after long filter runtimes. Further study of carbon attrition during extraction is recommended.

1 | Introduction

Recent advances in molecular technology have enabled researchers to characterize and quantify microbial communities in diverse natural and engineered environments. Applying a molecular approach to biofilter characterization may lead to improvements in this water and wastewater treatment technology by providing insights into the microbial communities responsible for biodegradation, the primary driver of contaminant removal. To ensure accuracy in this application, bacterial cells must be collected in sufficient concentrations from these environments without compromising sample quality. For example, flow cytometry (FCM) requires cell detachment from media to create single-particle suspensions without impacting membrane integrity or cell morphology. Molecular applications, such as quantitative PCR (qPCR) or digital PCR (dPCR), amplicon-based

sequencing, metagenomics, and quantitative stable isotope probing (qSIP), require sufficient concentrations of high-quality nucleic acids. Incomplete removal of the biofilm or destruction of cell and nucleic acid integrity could bias or limit subsequent microbial community analysis. Therefore, it is critically important to develop methods for biofilm removal that maximize biomass recovery and preservation.

While bacterial cell isolation has been investigated extensively for environmental water and soil matrices (Cruz-Cano et al. 2024; El Mujtar et al. 2022; Khalili et al. 2019; Steffan et al. 1988), substantial challenges remain when isolating cells from biofilms formed within biofilters in drinking water and water reuse applications. When performing cellular viability assessments with FCM, for example, biomass is often subjected to low-frequency ultrasonication (20–100 kHz) to generate the

Article Impact Statement

When studying biofilter microbial communities, it is crucial to capture sufficient biomass. Filter media type, age, and operational status at sample collection impact the yield of intact cells and DNA.

single-particle suspensions required for proper FCM analysis (Abzazou et al. 2015; Foladori et al. 2010, 2007; Frossard et al. 2016). However, results vary when evaluating sonication protocols for their ability to maximize intact cell yields, as evidenced by studies of activated sludge. Foladori et al. (2010) sonicated activated sludge samples at 20 kHz to disaggregate flocs prior to staining for intact and total cell counts, while Abzazou et al. (2015) found that sonication at 50/60 kHz, specifically nine cycles of 45 s, resulted in the highest recovery of viable cells. Frossard et al. (2016) used three 20-s durations at 20 kHz, followed by 20-s rests, to release cells from soil, sediment, and sludge samples. However, they found that FCM cell counts were lower than counts based on epifluorescence microscopy (EM), though it was unknown whether this was due to an overestimation by EM or an underestimation by FCM. Conversely, other studies found that high-frequency sonication (> 100 kHz) led to better results. For instance, Joyce et al. (2011) found that lower sonication frequencies (≤ 40 kHz) caused a decrease in cell viability, while a higher frequency (580 kHz) broke up clumps of cells without lysing cell membranes. One reason for this may be due to the larger cavitation bubbles that occur at lower frequencies, which then collapse with greater energy (Ashokkumar et al. 2010). However, the sonication experiments in Joyce et al. (2011) were carried out in suspended pure cultures, which may not be representative of the complex biomass suspensions recovered from a granular media filter.

Few studies have outlined effective protocols for removing intact cells from granular media, and of these existing studies, the focus has primarily been on sand. For instance, Vignola et al. (2018) found that high-energy sonication (i.e., using an ultrasonic probe) was more effective than low-energy sonication at removing biofilms from sand biofilter particles, with the highest microbial count resulting from four stepwise increases in high-energy sonication. The use of dispersants—often in the form of surfactants that promote particle release and prevent aggregation—has also yielded mixed results. Vignola et al. (2018) found that a solution of polysorbate 80 (or Tween 80) and the ionic dispersant sodium pyrophosphate maximized total cell counts from filter media, while intact cell counts were maximized when using autoclaved tap water as the dispersant. Thus, preferred approaches may differ depending on the ultimate goal (e.g., maximizing intact vs. total cell counts).

DNA extraction from biofilter media can also pose challenges, especially for aged granular activated carbon (GAC) (i.e., biological activated carbon [BAC]) (Bacaro et al. 2019). Commercially available kits have been designed for diverse environmental matrices, including biofilms, soil, and water samples, and involve a cell lysis step, which can be either chemical, physical, or a combination of both (Danaeifar 2022). Most kits designed to

process “difficult samples” (i.e., complex matrices such as soils and biofilms) will typically include chemical/enzymatic lysis followed by physical lysis (bead-beating) to efficiently disrupt cell membranes and release microbial nucleic acids. However, it is also important to minimize nucleic acid shearing and damage (Liesack et al. 1991), and even damage of sample constituents (e.g., GAC itself) that might lead to downstream interference. The bead-beating step has been extensively tested (Cullen and Hirsch 1998; de Boer et al. 2010; Salonen et al. 2010), providing researchers with a variety of choices in terms of bead material (e.g., glass, ceramic, metal, or garnet) and size. There are general recommendations for bead-beating times and intensities included in product guides and instructions for commercial kits, but these parameters should be optimized for each study to ensure that abundant high-quality DNA or RNA is obtained. Failing to obtain high-quality nucleic acids will inevitably compromise all downstream processes, leading to biased findings and conclusions (Feinstein et al. 2009; Roopnarain et al. 2017).

Recovering and preserving biofilm components as effectively as possible is the first step in accurately quantifying and characterizing microbial communities and tracking microbial evolution within these complex environments. To this end, the goal of this study was to improve biofilter media sample collection and processing, specifically to maximize recovery and preservation of intact cells for FCM analysis and high-quality DNA for molecular applications. We focused on sonication duration at low frequency, DNA extraction kit selection and bead-beating modification, and media collection timing with respect to backwashing as the key parameters to maximize intact cell and DNA yields. Biofilter media included anthracite and GAC of varying age.

2 | Methodology

2.1 | Biofilter Characteristics and Sampling Technique

This study was performed using granular media from two different biofiltration facilities located in North America. Table 1 summarizes the properties of the biofilters and biofilter media at both facilities, and Table 2 summarizes the experimental goals and conditions. The effects of GAC media age (20+ years old or < 1 year old) and filter operational status (runtime of zero hours for post-backwash samples and typically 20–30 h for pre-backwash samples) were evaluated using samples collected from full-scale drinking water biofilters located at the David L. Tippin (DLT) Water Treatment Facility in Tampa, Florida, USA. It was hypothesized that older media collected prior to backwashing would yield greater cell counts and higher DNA concentrations compared to new media. The DLT facility treats approximately 310,403 m³/day of primarily surface water, with up to 10% groundwater during the dry season (typically April–August). At this facility, water is first screened to remove large debris, followed by coagulation/flocculation/sedimentation, ozonation, biofiltration, and chlorine disinfection. We selected four GAC biofilters that we refer to as filters 1–4 (F1–F4). At the time of this study, the range of average hydraulic loading rates for all filters at this facility was 117,176 m³ m⁻² d⁻¹ (2–3 gpm ft.⁻²), and the average backwash

TABLE 1 | Properties of virgin biofilter media, media age, and bed depth in study biofilters.

Filter bed	Drinking water				Water reuse	
	F1	F2	F3	F4	C1	C2
Age at time of study*	20+ years	20+ years	12 months	8 months	10+ years	20+ years
Type	GAC	GAC	GAC	GAC	Anthracite, reagglomerated bituminous coal	GAC, bituminous coal
Product	GAC 830	GAC 830	GAC 820	GAC 820	Unknown	GAC 820
Manufacturer	Unknown	Unknown	Carbon activated corporation	Carbon activated corporation	Leopold engineered filter media	Cabot corporation
Effective size (mm)	1.12	1.12	1.00–1.20	1.00–1.20	1.35–1.46	0.86
Uniformity coefficient	1.5	1.5	1.6	1.6	<1.4	2
Bed depth for carbon (cm)	56	56	64	64	165	165
Bed depth for sand (cm)	30	30	23	23	30	30

*The exact replacement dates/ages for columns C1 and C2 and filters F1 and F2 are unknown, except that the media had been in operation for 10–20+ years at the time of collection.

TABLE 2 | Biofilter media types and runtimes at the time of sample collection for each phase of the study. Details related to each full-scale drinking water filter (F) and pilot-scale reuse column (C) are provided in Table 1.

Filter description	Media type	Runtime upon sample collection (hours)
Phase 1. Impact of sonication time on FCM analysis		
Drinking water F1	GAC	28
Drinking water F3	GAC	26
Phase 2. Impact of kit modifications on DNA extraction efficiency		
Drinking water F1	GAC	28
Drinking water F3	GAC	26
Drinking water F4	GAC	27
Water reuse C1	Anthracite	~0 ^a
Water reuse C2	GAC	~0 ^a
Phase 3. Impact of biofilter media age and operational status on FCM analysis and DNA extraction efficiency		
FCM analysis		
Drinking water F2	GAC	18
Drinking water F3	GAC	26
DNA extraction		
Drinking water F1	GAC	28
Drinking water F3	GAC	23

^aSamples collected immediately post-backwash.

rates were $344 \pm 1.8 \text{ m}^3 \text{ m}^{-2} \text{ d}^{-1}$ ($5.86 \pm 0.03 \text{ gpm ft.}^{-2}$) during low-rate backwash and $851 \text{ m}^3 \text{ m}^{-2} \text{ d}^{-1}$ ($14.4 \pm 0.07 \text{ gpm ft.}^{-2}$) during high-rate backwash. The average water temperature

was approximately 25°C. Sampling was performed using a filter bed core sampler (32 mm inner diameter) inserted approximately 20 cm into the filter bed. The top 15 cm of GAC (approximately 120 mL) was collected from the core using a sterile spatula and placed into autoclaved sample jars.

To compare DNA extraction efficiency between anthracite and GAC, we utilized two pilot-scale biofilter columns located at the Clark County Water Reclamation District (CCWRD) in Las Vegas, NV, USA. The feed water for the pilot-scale biofilters was full-scale tertiary wastewater effluent supplemented with pilot-scale ozonation. The full-scale system treats approximately 380,000 m³/day of municipal wastewater starting with screening and grit removal, primary clarification, and activated sludge treatment with full nitrification, partial denitrification, and biological phosphorus removal. This is followed by secondary clarification, granular media filtration, and ultraviolet (UV) disinfection. Prior to UV disinfection, the full-scale tertiary effluent was diverted to a pilot-scale ozone system operating at a mass-based ozone to dissolved organic carbon ratio (O₃/DOC) of 0.8. The ozonated tertiary effluent was then fed to pilot-scale biofilter columns operating in parallel, one of which was anthracite (C1) and the other was GAC (C2). The hydraulic loading rate was approximately 217 m³ m⁻² d⁻¹ (3.7 gpm ft.⁻²), and the backwash rate was approximately 897 m³ m⁻² d⁻¹ (15.3 gpm ft.⁻²). Using a sterile spatula, approximately 500 g of filter media were collected through media sample ports located at a depth of approximately 20 cm from the surface of each of the partially drained columns and placed into sterile bottles.

2.2 | Sonication

To determine the most efficient sonication time that balanced increasing biofilm removal for GAC, decreasing GAC breakage, and decreasing cell lysis, a series of sonication interval and intensity experiments were conducted on drinking water treatment facility GAC of varying ages. Media was collected

from filters F1 (20+ year old GAC) and F3 (1-year-old GAC) (Table 1) on the same day after runtimes of 28 and 26 h, respectively (Table 2). Triplicate GAC samples of 1 g wet weight were suspended in 10 mL of Milli-Q water and sonicated at 40 kHz (i.e., low frequency) using a Branson M2800 Ultrasonic Cleaner (Branson Ultrasonics, Brookfield, Connecticut, USA). The sonication energy density was calculated as follows:

$$\text{SED} = \frac{P_U t}{V}$$

where SED is the sonication energy density (Ws/mL), P_U is the ultrasonic power delivered (W), t is the duration for which power is applied (seconds), and V is the volume of water in the sonication tank (mL). This ultrasonic cleaner model has a tank volume of 2.81 L (0.00281 m³) and maximum ultrasonic power of 110 W. Sonication times ranged from 1 to 40 min, corresponding to SED inputs of ~2.3–94.0 Ws/mL; see Figure S1 for a graph of sonication time and corresponding SED for this study. To identify the effective range of applicable sonication times, the F3 filter containing new carbon was sonicated for up to 40 min, collecting samples in 5-min (11.7 Ws/mL) intervals. To increase resolution and identify the impact of GAC age, samples from F1 (20+ years old) and F3 (1 year old) were sonicated in 1-min (2.3 Ws/mL) increments up to 15 min (35.2 Ws/mL). After sonication, the sample was mixed using a pipette to achieve homogenization prior to sample collection for subsequent FCM analysis.

Cell classification by FCM was accomplished using an intact/lysed cell staining protocol, specifically by staining two aliquots of each sample in Falcon flat-bottom 96 well plates (Corning Inc., Corning, NY). The first aliquot was stained with SYBR Green I (SGI) (10,000×) (Invitrogen, Waltham, Massachusetts, USA) diluted in 0.22-μm-filtered dimethyl sulfoxide (DMSO) to achieve a working stock concentration of 100×; 2 μL of working stock was then added to 198 μL of sample to achieve a final SGI concentration of 1×. The second aliquot was stained with both SGI and propidium iodide (PI) (Invitrogen) diluted in filtered DMSO to achieve a working stock concentration of 100× SGI and 40 mM PI. Then, 2 μL of this working stock was added to 198 μL of sample for a final concentration of 1× SGI and 0.4 mM PI. Each sample totaled 200 μL after staining. This intact/lysed cell staining protocol red-shifts lysed cells due to the PI counterstain (Gatza et al. 2013; Props et al. 2016). After staining and incubation in the dark for 10 min at 35°C, cell counts were performed with a BD Accuri C6 Plus Flow Cytometer (BD Biosciences, Erembodegem, Belgium) equipped with a CSampler Plus automated sample loader and using Milli-Q water as sheath fluid. The instrument was set to a volumetric run limit of 50 μL at fast speed using an FL1-H acquisition/signal intensity threshold of 350. Quality control was performed in the software daily using BD CS&T RUO beads, containing a known median fluorescence intensity and distribution. After exporting raw FCM data, cell signals were denoised from instrument signals and debris using a virtual gate, which was applied using R programming (R Core Team 2022), Phenoflow (Props et al. 2016), flowCore (Ellis et al. 2017b), and flowViz (Ellis et al. 2017a). With SGI staining, events (i.e., particle signals) within the gate were identified as “total” cells. Upon counter-staining the

second sample aliquot with PI, damaged cells and debris were red-shifted outside the gate, excluding them from the “intact” cell count within the gate. “Damaged” cell counts were determined as the calculated difference between the “total” and “intact” cell counts.

2.3 | DNA Extraction Efficiency

In evaluating DNA extraction efficiency, we tested media type, media age, and filter operational status. Media samples were collected from the full-scale drinking water treatment facility (F1, F3, F4) and the pilot-scale reuse system (C1, C2) (see Table 1 for sample characteristics and Table 2 for runtimes at sample collection). Media samples were stored at 4°C within 1 h of collection. At least one sample of Milli-Q water was included in every DNA extraction as a negative extraction control. DNA extracts were frozen immediately at –20°C. To quantify DNA in each extract, fluorometric quantification with a Qubit 4 Fluorometer (Thermo Fisher Scientific Inc., Waltham, MA) and the Qubit double-stranded DNA (dsDNA) HS (High Sensitivity) Assay kit was selected, as this method is more reliable, sensitive, and accurate than spectrophotometry-based methods such as NanoDrop (Bacaro et al. 2019; Invitrogen 2018), particularly at low concentrations (Billington et al. 2023).

Extraction kits included the DNeasy PowerSoil Pro Kit and the DNeasy PowerMax Soil Kit (QIAGEN) to compare extraction efficiency between media type, media age, and filter operational status. Additionally, the DNeasy PowerBiofilm Kit and the Quick-DNA Fecal/Soil Microbe Midiprep Kit (Zymo Research Corp, Irvine, CA) were included for extracting the pilot-scale reuse media to evaluate a wider range of cell lysis methodologies, including a heating step and a variation in bead type. The variations identified in each kit are shown in Table 3. It was not possible to test all DNA extraction kits and modifications on all filters listed in Table 1. Specifically, the collection of GAC media from the full-scale filter beds (F1–F4) at the drinking water treatment facility required careful planning and coordination with facility operators. Samples from the drinking water treatment facility were only tested with the DNeasy PowerSoil Pro Kit and the DNeasy PowerMax Soil Kit (QIAGEN) due to their availability at the time of sampling at that site. Additionally, collecting media from a specific filter at a predetermined runtime (or immediately post-backwash) was particularly challenging, as only one filter at a time is taken offline for backwashing. Also, only 2 of 30 biofilter beds contained GAC with an age of < 20 years at the time of this study, which further limited our ability to collect samples satisfying specific criteria.

From prior work with GAC for subsequent FCM analysis (described later in Section 3.1), longer sonication duration led to a decrease in intact cells harvested from media samples, so it was hypothesized that reduced vortex action (less vigorous) or shorter extraction methods would yield more DNA per gram of media (i.e., increased DNA extraction efficiency). Note that for all three QIAGEN kits, the protocols advise using a horizontal vortex adapter and vortexing at maximum speed for 10 min, while the Zymo kit recommends optimizing the speed and

TABLE 3 | DNA extraction kits, specifications, and modifications.

Extraction kit	Media quantity, wet (g)	Final elution volume (μL)	Bead composition	Bead size (mm)	Cell lysis procedure	Modifications and filters tested for optimization experiments			
						No modification ^a	Bead tube inversion	Slow-speed vortex ^b	Reduced-duration vortex ^c
Qiagen DNeasy PowerBiofilm Kit	0.2	100	Glass and ceramic	0.1–2.4	Heat at 65°C, bead beat	C1, C2	C1, C2	C1, C2	C1, C2
Qiagen DNeasy PowerSoil Pro Kit	0.2	100	Garnet, metal, glass, and ceramic	0.1–2.8	Bead beat	C1, C2, F1, F3, F4	C2, F4	C2, F4	—
Qiagen DNeasy PowerMax Soil Kit	10	5000	Garnet	0.7	Bead beat	C1, C2, F4	C2, F4	C2	—
Zymo Research <i>Quick-DNA</i> Fecal/Soil Microbe Midiprep	2.5	600	Silica-based/polypropylene	0.5	Bead beat	—	C1, C2	—	C1, C2

^aVortex speed set to maximum and vortex duration of 10 min.

^bVortex speed less than maximum setting = “slow.”

^cVortex duration < 10 min = “reduced.”

duration for complete lysis and does not specify these parameters. Therefore, modifications to QIAGEN's extraction methods included reduced bead-beating speed (from max speed of 3200 rpm to the lowest speed setting of 600 rpm) and reduced duration (from 10 min down to 4 min, 1 min, or 30 s), or skipping the bead-beating step altogether and inverting the tubes manually. An additional modification avoided all vortex steps until GAC had been separated from the supernatant.

3 | Results and Discussion

3.1 | Determining Sonication Time for FCM Analysis

The highest intact and total cell counts were observed at sonication durations of 10 min (23.5 Ws/mL) and 15 min (35.2 Ws/mL), respectively (Table 4). Beyond 10–15 min, intact and total cell counts either plateaued or demonstrated considerable variability between samples, perhaps due to the visually apparent degradation of GAC for durations ≥ 15 min (Figure 1). Based on the visual properties of the samples, it was hypothesized that sonication for ≥ 15 min (≥ 35.2 Ws/mL) caused fracturing of the GAC particles, potentially exposing additional adsorption sites for cellular reattachment. This helps explain the low cell counts observed at 20 and 40 min, for example. At durations ≥ 25 min (≥ 58.7 Ws/mL), there was less suspended GAC apparent in the samples, although there was still discoloration. It is hypothesized that this apparent reduction in suspended GAC is due to a combination of agglomeration and settling (i.e., grains clumping and settling) or additional decreases in the size of grains remaining in suspension. Reducing GAC breakage is important because GAC decomposition induced by sonication might interfere with subsequent analyses that rely on optical properties and/or reliable separation of target analytes.

This production of carbon fines during sonication of GAC has been observed in other work and was shown to increase linearly for a range of SEDs (0–400 Ws/mL), while recovery of DNA was strongly reduced at SEDs > 67 Ws/mL (Kim et al. 2014). The authors speculated that DNA adsorption to the carbon fines generated during sonication adversely impacted DNA recovery. Thus, limiting the degradation of GAC during ultrasonication may prevent the adsorption of cells to carbon fines and allow for greater biomass and DNA recovery for downstream analyses.

To further investigate the effect of sonication duration on intact cell counts given varying carbon ages, GAC aliquots from filters F1 (20+ years old) and F3 (1 year old) were sonicated in 1-min (2.3 Ws/mL) increments up to 15 min (35.2 Ws/mL) (Table 4) based on the initial 40-min study (Figure 1). The ideal sonication time was identified as the test condition resulting in the highest intact cell count prior to any plateau (i.e., point of diminishing return) or steep decline (i.e., excessive sonication). The older GAC from filter F1 yielded an ideal sonication time of 8 min (18.8 Ws/mL) (Figure 2A). The newer GAC from filter F3 exhibited the same sonication time for maximum intact cells as the older GAC (8 min or 18.8 Ws/mL) (Figure 2B), which is consistent with the ideal duration identified for the newer GAC in

TABLE 4 | Total and intact cell results for each GAC age group, sonication duration, and corresponding sonication energy density (SED).

Filter description	Sonication duration tested		SED tested		Point of maximum ^a intact cells		Point of maximum total cells		Point of sharp decline in total cells	
	Increments (min)	Range (min)	Increments of SED (Ws/mL)	Range of SED (Ws/mL)	Duration at maximum (min)	SED at maximum (Ws/mL)	Duration at maximum (min)	SED at maximum (Ws/mL)	Duration at minimum (min)	SED at minimum (Ws/mL)
Newer GAC, F3	5	0–40	11.7	11.7–94.0	10	23.5	15	23.5	20	47.0
Newer GAC, F3	1	0–15	2.3	2.3–35.2	8	18.8	8	18.8	9	21.1
Older GAC, F1	1	0–15	2.3	2.3–35.2	8	18.8	8	18.8	10	23.5

^aPoint of maximum intact cells refers to the initial maximum of intact cells, or the point of diminishing return.

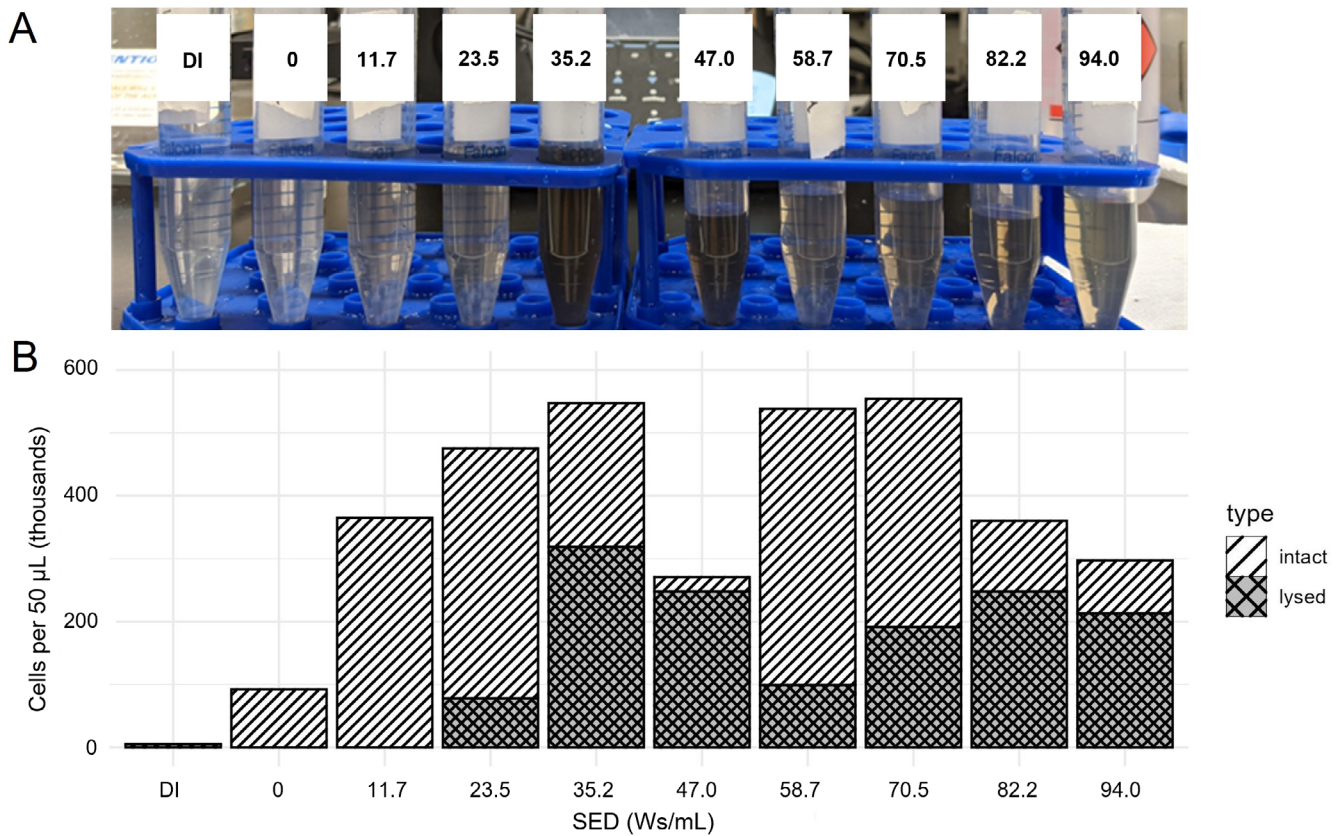


FIGURE 1 | Intact and total (intact + lysed) cell counts via flow cytometry with intact/lysed cell staining for ‘new’ GAC media from filter F3 (1 year old). (A) visual sample quality and (B) cell counts are shown as a function of SED (0–40 min, corresponding to 0–94 Ws/mL). Deionized (DI) water is included as a background control.

the first sonication experiment (Figure 1). However, we identified 7 min (SED of 16.4 Ws/mL) as the ideal sonication duration, specifically a duration that (1) yields a maximum or near-maximum intact cell count and (2) provides a buffer against the sharp decline that occurs at 9 min (Figure 2B). Similarly, extended sonication durations resulted in GAC breakage and intermittent drops in intact cell counts for both the older (F1) and newer (F3) GAC. Also, the intact cell counts for the newer GAC appeared to be systematically higher than for the older GAC. Interestingly, Figures 1 and 2 both show a bimodal distribution for cell counts for both older GAC (Figure 2A) and newer GAC (Figures 1B and 2B), as the total and intact cell counts begin to increase again after the initial increase and sharp decline. It is hypothesized that further sonication beyond the steep decline is due to the re-release of cells that had attached to the fractured carbon, though this should be explored further to confirm this hypothesis.

These results indicate that low frequency sonication (40 kHz) can yield total and intact cell counts that vary with sonication duration (i.e., energy input) and the age of the sampled filter media. Notably, newer GAC yielded greater intact cell counts overall, even without sonication (a sonication duration of zero minutes). One hypothesis is that the use of highly abrasive air-scour during backwash for 20+ years caused a decrease in the effective size and an increase in the uniformity coefficient of the older GAC (Grens and Werth 2001). The difference in GAC grain sizes between older and newer GAC from

the DLT facility was confirmed with microscopic imaging (see Supporting Information section Text S1). Select GAC grains from F3 (new GAC) had lengths of approximately 0.6–1.3 mm (Figure S2), compared to the shorter lengths (<0.1 mm) of the GAC grains from F2 (old GAC) (Figure S3). Microscopic imaging was also performed to visualize the impact of sonication on the GAC grains from F3 (new GAC), including the production of carbon fines. Representative grains are visualized and characterized before sonication (Figure S2), after the ideal sonication duration of 7 min (SED of 16.4 Ws/mL) (Figure S3), and after 9 min of sonication (SED of 21.1 Ws/mL) (Figure S4). The appearance of GAC grains with lengths <0.1 mm confirmed the fracturing effect of sonication.

Manual and mechanical grinding have been shown to remove the outer layers of GAC grains. As radial pore volume decreases toward the grain core, grinding may reveal the less porous core, thus reducing the specific surface area of larger grains and producing smaller grain fragments with a higher specific surface area (i.e., redistributing porosity across size classes) (Abulikemu et al. 2023). The higher specific surface area of new GAC grains, along with its irregular surface, can promote biomass development by offering protection from shear stress (Pharand et al. 2014; Urfer et al. 1997). This benefit presumably declines as the surface properties change over time. These differences highlight the potential value of system-specific optimization of sonication conditions for laboratory analyses.

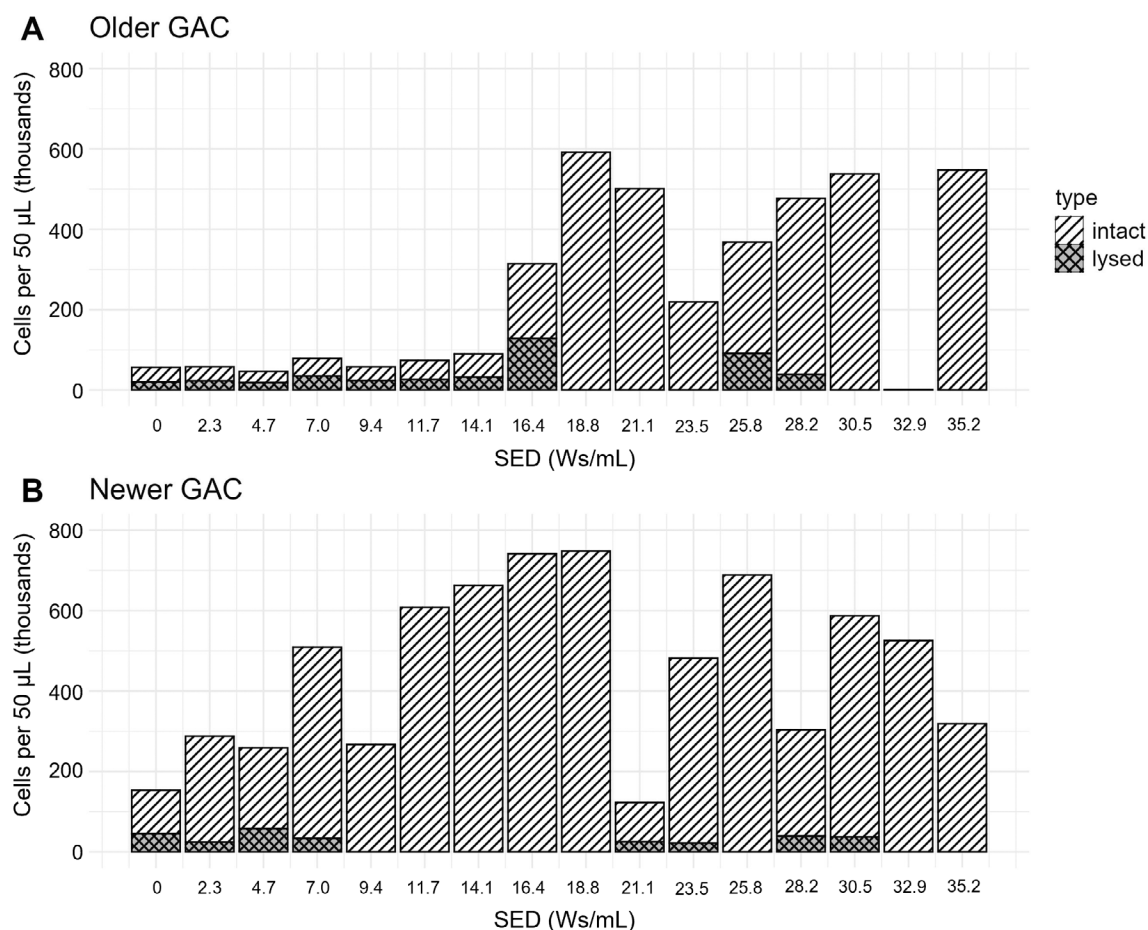


FIGURE 2 | Comparison of intact cell counts per 50 µL of GAC eluate for different filter ages as a function of SED. (A) Older (20+ years old) GAC from filter F1 at 28 h runtime and (B) newer (1 year old) GAC from filter F3 at 26 h runtime. Sonication duration was increased in 1-min (2.3 Ws/mL) increments up to 15 min (35.2 Ws/mL). Duplicate media samples were analyzed for each minute of sonication, with bars representing the average of replicates.

3.2 | DNA Extraction

Comparing the two reuse columns, one with anthracite (C1) and one with GAC (C2), there were noticeable differences in DNA yields. On average, column C1 with anthracite yielded 10-fold higher DNA concentrations across all DNA extractions performed ($16.2 \pm 10.4 \text{ ng}/\mu\text{L}$) in comparison to column C2 with GAC ($1.6 \pm 3.9 \text{ ng}/\mu\text{L}$). However, this discrepancy does not appear to be driven by differences in biomass density. Adenosine triphosphate (ATP) concentrations indicate the level of microbial activity in a water sample or on a surface and are often used as a measure of biomass abundance. Studies indicate ATP concentrations are either similar for GAC and anthracite (10^2 – $10^3 \text{ ng ATP}/\text{cm}^3$ media) (Arnold et al. 2018; Pharand et al. 2014), or potentially higher for GAC in some cases (de Vera et al. 2018; Bacaro et al. 2019). Thus, assuming equal extraction efficiency, DNA yields should be similar or higher for GAC, although there may be differences in GAC yield as a function of age. ATP was not measured in the current study, as microbial activity was assessed using cell counts via FCM. As noted in the previous FCM analysis, the fact that the GAC was 20+ years old and had likely degraded over time may have contributed to its low DNA yield. The anthracite in reuse column C1 was also 10+ years old at the time of this

study, but relative to GAC, anthracite is not expected to experience the same level of degradation over time due to its material properties (Crittenden et al. 2012).

When comparing across DNA extraction kits without modification (Figure 3A), anthracite yielded similar DNA concentrations for each kit, suggesting that the biofilm on this type of medium was efficiently and/or consistently removed despite the varying bead sizes and compositions. For Figure 3A, baseline conditions with no kit modifications are shown with squares, and modified conditions are shown with diamonds. For Figure 3B, modifications are shown in order of most aggressive bead-beating (modification = “none”) to least aggressive bead-beating (“invert”); the data are the same as in Figure 3A but reorganized to illustrate the impact of bead-beating modifications. After incorporating extraction kit modifications, DNA yields from anthracite were maximized with the longest and most intense vortexing conditions, regardless of which kit was used (Figure 3B). Resulting DNA concentrations did not differ significantly between delayed vortexing and the tube inversion approach (i.e., no vortexing), so these results were combined. Reuse column C2 with GAC yielded much lower DNA concentrations than anthracite when following extraction procedures from the manufacturers. The

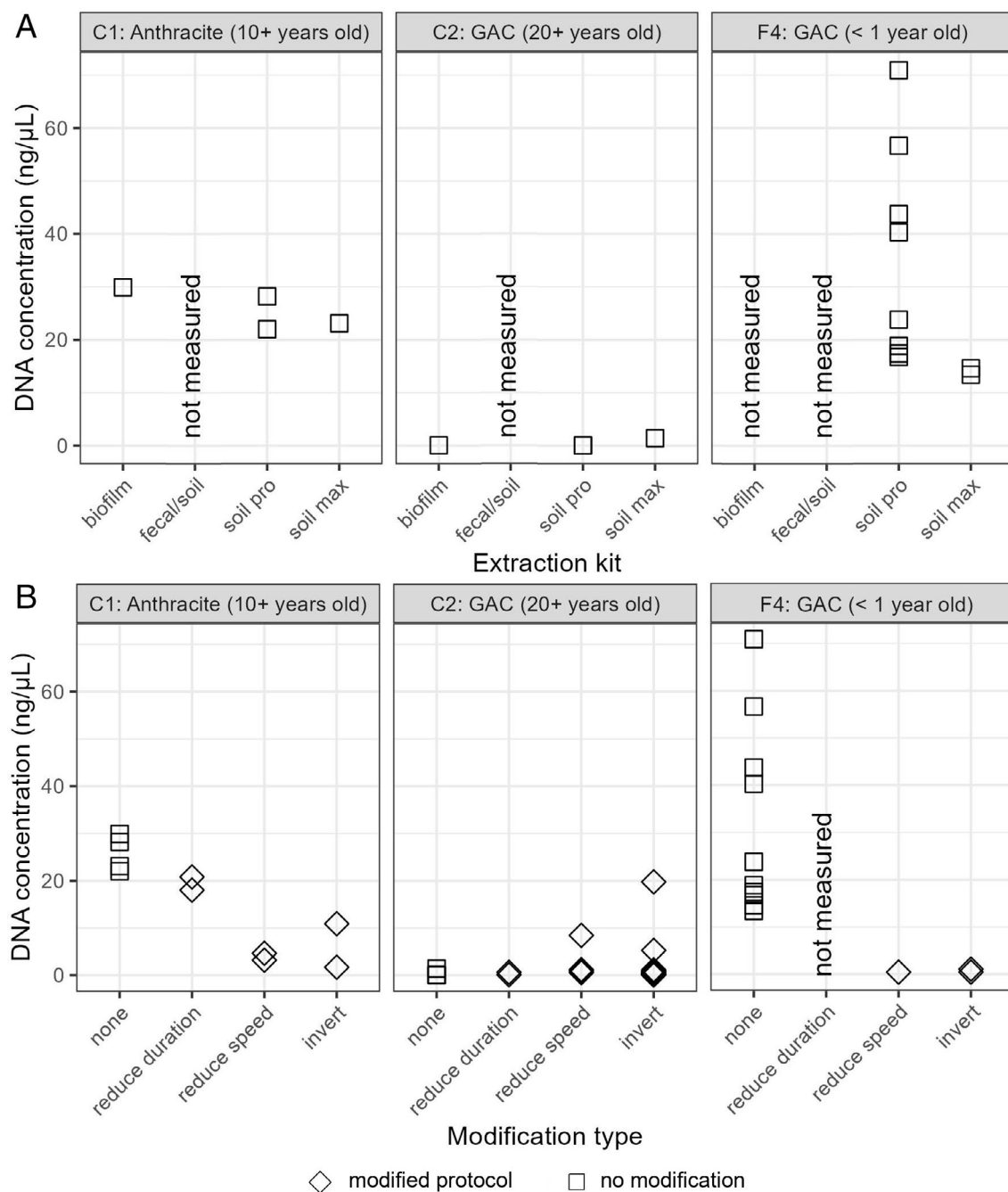


FIGURE 3 | Resulting DNA concentrations when extracting from 1 g (wet weight) of biofilter media using (A) various extraction kits without modification and (B) with various modifications. Squares represent results without modification to the kit protocol, while diamonds represent results of modifications. Kits tested included QIAGEN DNeasy PowerBiofilm Kit (“biofilm”), Zymo Research Quick-DNA Fecal/Soil Microbe MidiPrep (“fecal/soil”), QIAGEN DNeasy PowerSoil Pro Kit (“soil pro”), and QIAGEN DNeasy PowerMax Soil (“soil max”). No bead-beating protocol is specified in the fecal/soil kit product guide.

best-performing kit for the 20+ year-old GAC samples was the PowerMax Soil kit, which uses a medium-sized garnet bead and a larger quantity of media in each extraction (10 g vs. 0.2–2.5 g) (Table 3). In contrast with anthracite and GAC <1 year old, DNA yields for 20+ year-old GAC were highest when kits were modified by removing aggressive vortexing steps (Figure 3B), including omitting bead beating altogether. The effects of GAC age on biomass yield are covered more extensively in Section 3.3.

Hardness of biofilter media directly impacts its ability to withstand repeated backwashing, which is particularly relevant in this study since the filters had already been subjected to many backwash cycles over one to two decades of operation. Both anthracite and GAC are on the low end of the Mohs hardness ranking scale compared to sand: anthracite ranks at 2.3, sand at 7, and GAC is simply classified as “Low” (Crittenden et al. 2012). This propensity to break down in filter beds as a result of abrasion during backwash is well-documented (Grens

TABLE 5 | Summary of DNA extraction results for QIAGEN Dneasy PowerSoil Pro kit without modification. A runtime of 0h represents samples collected immediately after backwashing. Triplicate 0.25 g GAC samples were extracted for each filter and runtime. Data from filters F1 and F3 are also included in Figure 5B.

Filter description	Media type	Runtime at collection (hours) ^a	Media age at time of study	(DNA) (ng/μL)
Water reuse column C1	Anthracite	0 (post)	10+ years	25.1 ± 3.1
Water reuse column C2	GAC	0 (post)	20+ years	0.043 ± 0.008
Drinking water F1	GAC	28 (pre)	20+ years	0.089 ± 0.005
Drinking water F1	GAC	0 (post)	20+ years	20.4 ± 1.6
Drinking water F3	GAC	23 (pre)	12 months	28.1 ± 11.4
Drinking water F3	GAC	0 (post)	12 months	56.0 ± 12.5

^aIndicates whether sample was collected pre-backwash or post-backwash.

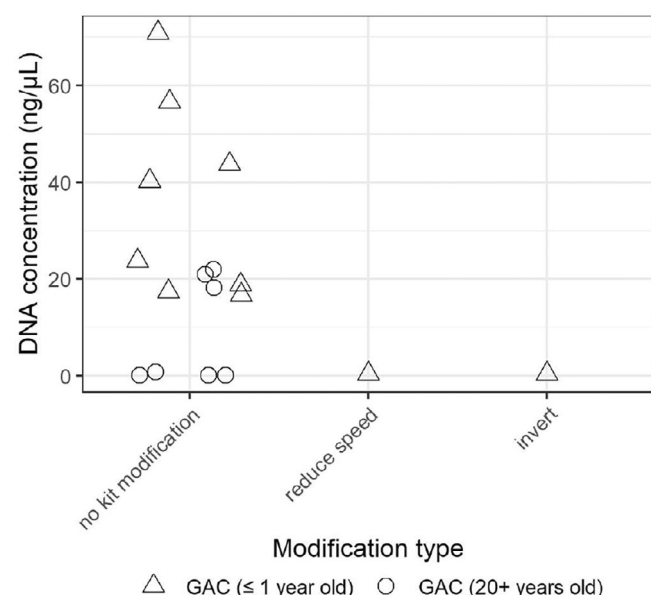


FIGURE 4 | Concentrations of DNA from drinking water biofilters F1 (20+ years old) and F3-F4 (≤1 year old). The extraction kit was QIAGEN Dneasy PowerSoil Pro, with modifications shown for new filter media and a comparison of media ages for the kit without modification.

and Werth 2001), though specific impacts have not been investigated beyond the decrease in media effective size. Because anthracite is more resistant to attrition than GAC, this characteristic would be expected to have impacts in treatment applications (e.g., decomposition during repeated backwashes) and laboratory applications (e.g., bead-beating). The full-scale facility (DLT) periodically adds virgin GAC to address decreases in bed depth due to attrition and losses from backwashing.

DLT also reports carbon fines clogging filter underdrains and air-scour nozzles as the main issue leading to the decision to replace the 20+-year-old GAC. It is hypothesized that GAC breakage creates new adsorption sites for dissolved organic molecules such as DNA, thereby reducing extraction efficiency (Yapsakli et al. 2010). Carbon fines are also known to reduce DNA extraction efficiency, and researchers speculate that the

main mechanism responsible is the adsorption of DNA to carbon fines during extraction (Kim et al. 2014). It is hypothesized that low-frequency sonication results in the breakdown of older carbon at extended sonication times, as discussed previously in Section 3.1, which may open up new interior sites for cell attachment. It is similarly hypothesized that the lower DNA concentrations observed in older biofilter GAC could be a result of the carbon breaking apart during the bead-beating process and exposing internal pores in the GAC; a study involving biochar found that increased porosity caused by high pyrolysis temperatures was associated with higher DNA adsorption capacity, resulting in lower DNA extraction efficiency (Dai et al. 2017; Wang et al. 2014). While evidence in support of GAC's fragility with age is presented in the current work, future research should explore the exact mechanisms of GAC breakage and internal pore accessibility to determine the validity of this inference. Additionally, the age of GAC in the older filters included in this study (20+ years old) is not unusual, as the Water Research Foundation's Biofiltration Guidance Manual (Water Research Foundation 2020) notes that some biofilters have been using the same media for up to 30 years, so methods to improve biomass recovery from GAC should not neglect this media age group.

It is essential to extract sufficient quantities of high quality microbial DNA for downstream molecular applications. For example, reproducible amplification during library construction for amplicon-based sequencing requires DNA template concentrations of 5–10 ng/μL (Kennedy et al. 2014); lower initial DNA concentrations require additional PCR cycles, leading to proportionately greater chimera formation (D'Amore et al. 2016; Lahr and Katz 2009). Similar target DNA concentrations have been reported for qSIP to identify metabolically active bacterial taxa (Dunford and Neufeld 2010; Neufeld et al. 2007). Our results indicate that downstream molecular analyses would be feasible for nearly all anthracite samples, while DNA concentrations from 20+ year old GAC may not be sufficient unless the extraction protocol is modified. Successful microbial community analysis can be achieved at initial DNA concentrations as low as 0.04 ng/μL (Multinu et al. 2018) or even 0.001 ng/μL (Biesbroek et al. 2012), but excessively low template concentrations may magnify impacts from background contamination (Biesbroek et al. 2012). Thus, it is critically important to maximize DNA yields for these downstream analyses, particularly in

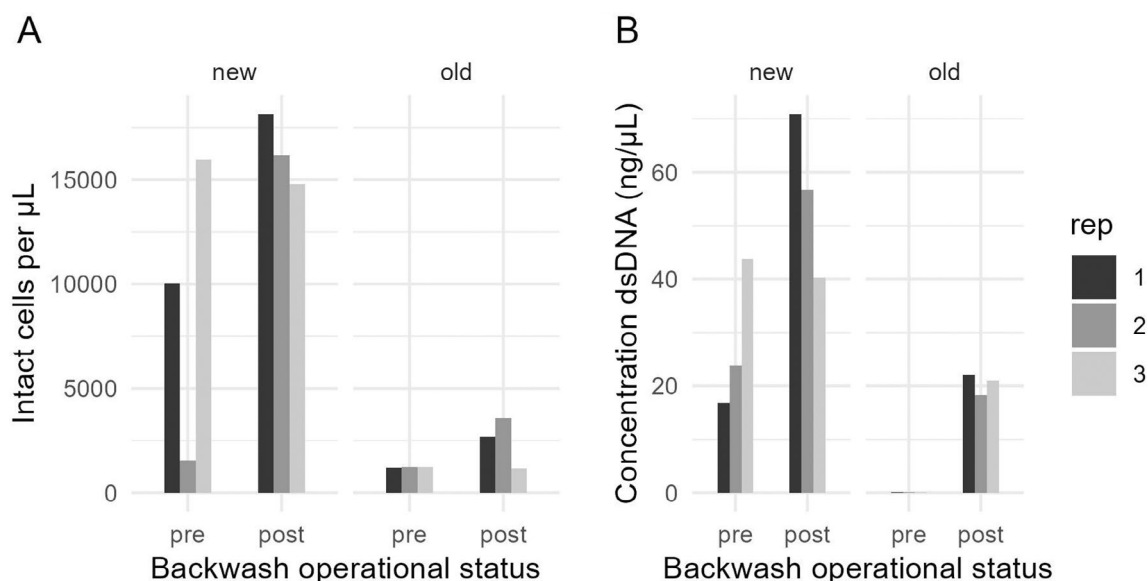


FIGURE 5 | (A) Intact cell counts from FCM analysis pre- and post-backwash for newly-replaced media from F3 (26 h runtime) and old media from F2 (18 h runtime). Replicates are shown for each filter and status. All samples were sonicated for 10 min (SED of 23.5 Ws/mL). (B) Concentration of DNA extracted pre- and post-backwash from filter F3 (new media, 23 h runtime) and filter F1 (old media, 28 h runtime) using the QIAGEN Dneasy PowerSoil Pro kit without protocol modification. Replicates are shown for each filter and status.

environments where biomass is already sparse. This again highlights the value of performing preliminary method optimization experiments to ensure the integrity of the results generated by a given study.

Similar to GAC from the water reuse biofilter, old GAC (20+ years old) from drinking water filter F1 yielded less DNA than new GAC (≤ 1 year old), shown in both Table 5 and Figure 4. It is estimated that both the water reuse GAC biofilter and drinking water filter F1 contained GAC over 20 years old, possibly contributing to the similarity in biomass yield from these two biofilters. Additionally, due to the likelihood of operational variation between the C2 GAC (water reuse) pilot column and F3 GAC (full-scale drinking water) biofilter during the life of the respective filter beds, it is not known whether the variation in DNA yield between these filters is related to the differences in organic load in these two separate treatment trains or to physical characteristics of the GAC (e.g., due to manufacturer specifications, filter operational decisions, or differences in media age). The newer GAC (≤ 1 year old) from filter F4 yielded exceptionally high amounts of DNA without modification of the extraction kit, with an average concentration of $36.1 \pm 20.3 \text{ ng}/\mu\text{L}$ (Figure 4). Studies of biofilter ATP have shown that biomass in anthracite biofilters reaches steady state after about 220 days of operation (Stoddart et al. 2016), while steady state is achieved in GAC biofilters after about 90 days (Velten et al. 2011). This suggests that all GAC biofilters included in this study should have reached a plateau in biomass, so differences in DNA concentrations between media types and across filter ages may not be attributed to ongoing acclimation of GAC ≤ 1 year old.

The DNA extraction process has proven challenging for complex environmental matrices—with DNA yields varying depending on sample origin and cell lysis, extraction, and purification protocols (Kuhn et al. 2017; Lever et al. 2015)—and has therefore been investigated relatively extensively

for water (Deiner et al. 2015; Djurhuus et al. 2017) and soil (Dai et al. 2022; Dineen et al. 2010; Pakpour et al. 2013) ecosystems. Many studies on the topic recommend in concluding remarks that researchers perform pilot extraction tests to ensure sufficient DNA can be collected from environmental samples for downstream molecular applications (Felczykowska et al. 2015; Lever et al. 2015), and the results we present here confirm this recommendation.

Overall, our recommendations for the use of DNA extraction kits to maximize DNA yield from biofilter media are to consider the age and type of media being processed. While we observed maximum DNA concentrations in new drinking water GAC (≤ 1 year old) when using a soil kit without modifying the manufacturer's protocol and with 0.25 g of media, old GAC (20+ years old) from reuse and drinking water systems greatly benefited from less-aggressive bead-beating methods, including inverting the tube instead of vortexing. Old GAC also benefited from the use of an extraction kit with a much greater mass of GAC per extraction tube, allowing for up to 10 g per extraction. Further research into aging GAC is required to determine whether this increase in DNA concentration is due to a limited quantity of biomass on old GAC compared to new GAC or to inefficient sample extraction. Pairing future DNA extraction studies with microscopy investigations of biofilm thickness and coverage could help illuminate the decreased DNA extraction from older GAC media.

3.3 | Biofilter Media Age and Operational Status

To determine an ideal media age and collection time in biofilter operation, we focused only on the full-scale biofilters of differing ages from a single facility to reduce variability. Our goal was to compare microbial biomass yields between GAC samples from the same water treatment facility prior to and after filter backwash, as

well as between both old (20+ years old) and newer GAC (≤ 1 year old). The hypothesis for these experiments was that media samples collected prior to backwashing would yield greater amounts of intact cells and DNA because the backwash procedure was expected to remove part of the biofilm. Prior to the study, we hypothesized that older GAC would yield more biomass than newer GAC due to the increased maturity of the biofilm; however, considering the more brittle material properties of older GAC, this was no longer expected. Figure 5 illustrates the effects of media age on intact cell counts and DNA recovery from corresponding pre- and post-backwash samples collected after 20–30 h of run time from the full-scale drinking water biofilters. The quantification of biofilm on older GAC resulted in lower biomass recovery, as evidenced by both intact cells and DNA concentration, regardless of water system and filter operational status at sample collection (Figure 5 and Table 5). This likely indicates an inadequacy in extraction technique when following identical protocols regardless of media age, as equal biomass would be expected on older GAC and newer GAC since the latter is likely to have reached steady state after the first 90 days of operation (Velten et al. 2011).

Based on our previous experiments, this older GAC is more likely to break apart during DNA extraction and result in decreased yield. The friability of older GAC used in this study is likely due to the use of air scour at the full-scale water treatment facility, in addition to media age, as studies of this abrasive cleaning process have shown up to a 3% loss in GAC bed depth per year (Humby and Fitzpatrick 1996; Pang et al. 2005). For reuse column C2, it is not known whether the original water reclamation facility employed air scour during backwash of the GAC that was later added to the pilot column.

When considering filter operation status at collection, the difference between pre- and post-backwash was less pronounced than with media age. We found greater intact cell counts quantified through FCM after filter backwash, compared to samples collected prior to backwash, for both new and old GAC, though there was considerable variability in post-backwash intact cell yields (Figure 5). DNA concentrations were consistently higher in post-backwash GAC media samples, with concentrations sometimes increasing from less than 1 ng/ μ L pre-backwash to over 20 ng/ μ L post-backwash (~200-fold increase for F1 and two-fold increase for F3, Table 5). It should be noted that this study does not characterize the structure and function of the microbial community in the pre-backwash versus post-backwash samples. Thus, our results do not clarify whether the higher cell counts and greater DNA quantities found post-backwash maintain the same microbial community characteristics as the pre-backwash samples. This should be confirmed with future study.

Several studies have investigated changes in biomass in the top layer of filter beds post-backwash, with some finding decreases in the phospholipid fatty acid concentration (Liao et al. 2015; Miltner et al. 1995) and others finding increases in ATP density after backwash (de Vera et al. 2018). Emelko et al. (2006) noted a significant decrease in phospholipid biomass only in cold water biofilters (1°C–3°C).

The increase observed in this study may be due to structural changes in the biofilm pre- and post-backwash. Before

backwashing, the GAC biofilm may be more structurally stable, allowing it to hold tightly onto the microbial community and extracellular matrix. When sonicating pre-backwash GAC, these biofilms may be more resilient and less prone to disruption by sonication, resulting in less DNA release. Backwashing can physically alter the biofilm structure, partially dislodging loose biofilms and cells (Niu et al. 2018). Our hypothesis is that post-backwash GAC biofilms, after being partially disturbed by backwash, might release more readily from GAC during sonication or DNA extraction, as they are in a more fragile state than the more robust biofilm found on pre-backwash GAC. In other words, while less biomass may be present overall after backwash, the biofilm that remains may be easier to remove with the methods presented in the current study. This hypothesis calls for further investigation.

Another potential explanation for the increase in DNA and intact cell recovery from GAC post-backwash is the presence of interfering compounds in the biofilter bed prior to backwash. Upstream of the drinking water biofilters, ferric sulfate is added for coagulation. Several authors have described patches of iron oxides on filter surfaces (Boersma et al. 2025; Corbera-Rubio et al. 2023; Gülay et al. 2018), and this was also observed at the DLT facility in the current study. Particulate iron that coats filter media has been shown to decrease post-backwash, as dislodged iron flocs are removed from the filter (Boersma et al. 2025). Metal ions are known to interfere with DNA extraction by binding to the negatively charged phosphate backbone of DNA (Anastassopoulou 2003), which is why most extraction kits include EDTA to chelate metal ions (Kuffel et al. 2021; Nkuna et al. 2022). However, the effectiveness of EDTA is highly dependent on metal ion concentration (Kuffel et al. 2021); thus, backwashing may reduce residual iron at the biofilter surface to levels at which EDTA is effective during DNA extraction from media. The effectiveness of EDTA is also limited by the inhibitory effects of the chemical itself, as Khosravinia and Ramesha (2007) demonstrated that concentrations of EDTA greater than 11 μ g/ μ L reduce DNA recovery.

These results are essential to the development of best practices for GAC media sampling as many downstream molecular applications might be hindered or confounded by the lower DNA yields from the pre-backwash samples. This study has shown that filter media age and operational status at collection are important considerations, as older GAC (20+ years old) sampled pre-backwash yielded very low concentrations of DNA (< 1 ng/ μ L) and intact cells ($< 100,000$ cells). Therefore, it is recommended that GAC from a newly-replaced filter bed (≤ 1 year old) be used for biofilter microbial analysis, as media age and total backwash cycles may reduce media durability and lead to breakage during sample processing. However, a suitable filter media age cutoff cannot be established solely based on this study, as no filter beds contained GAC between 1 and 20 years old. Further research is recommended to better characterize this threshold for both spent and virgin GAC.

4 | Conclusions

The effectiveness of biofiltration systems relies in large part on biofilms, and particularly on microbial activity within these

biofilms. Understanding the ecological interactions within these biofilms requires carefully designed and implemented sample collection, processing, and analysis. In this study, we investigated media type, filter media age, and filter backwash status, in addition to optimizing sonication and bead-beating conditions for FCM and molecular applications, respectively. Important findings include the following:

- Sonication duration exceeding 10 min (23.5 Ws/mL) at 40 kHz resulted in visual GAC breakage.
- Newer biofilter GAC yields higher intact cell counts and higher concentrations of DNA compared to old GAC.
- When working with older GAC, less-aggressive bead-beating (e.g., decreasing vortex speed and duration or avoiding vortexing altogether) improves DNA yields, and shorter sonication times (i.e., lower sonication energy densities) improve cell yields.
- Microbial DNA removal from anthracite biofilter media and from new GAC requires little modification to DNA extraction kits typically used in soil applications.
- Samples collected after longer filter runtimes yield less DNA and fewer cells from GAC, compared to media collected immediately after filter backwash.

These findings should be considered when characterizing biofilter microbial communities, as inadequate intact cell capture and DNA extraction may bias the data and lead to erroneous conclusions.

Author Contributions

Andrew Black: conceptualization, investigation, visualization, methodology, writing – original draft. **Natchaya Luangphairin:** conceptualization, investigation, methodology, writing – review and editing. **Katerina Papp:** conceptualization, supervision, methodology, writing – review and editing. **Daniel Gerrity:** conceptualization, supervision, methodology, writing – review and editing. **Eric Dickenson:** conceptualization, supervision, methodology. **Katherine Alfredo:** conceptualization, resources, supervision, methodology, writing – review and editing.

Acknowledgments

This work was partially supported by the National Science Foundation under Grant No. 1832713. The authors would like to acknowledge the efforts of the utility partners—the David L. Tippin Water Treatment Facility (City of Tampa Water Department) and the Clark County Water Reclamation District—for access to their facilities and for assistance with media sample collection. We would also like to acknowledge Dr. Jayme Peraud in the Harwood Lab at the University of South Florida for her DNA extraction assistance and Dr. Aarthi Mohan at the Southern Nevada Water Authority for her sample collection assistance.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1.**